

PIZZA SALES ANALYSIS USING SQL

A DATA ANALYST PROJECT

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This project demonstrates the use of SQL to analyze pizza sales data, extract meaningful insights, and support data-driven decision making



Key Skills Demonstrated:

- **SQL Queries & Joins**
- **Aggregations & Subqueries**
- **Revenue & Sales Analysis**
- **Business Insight Generation**





PROJECT OVERVIEW

The Pizza Sales SQL Project focuses on analyzing a structured sales dataset to understand customer behavior, product performance, and revenue trends

The analysis answers real-world business questions such as:

- **What is the total revenue generated?**
- **Which pizzas and categories perform best?**
- **Which pizza sizes contribute most to sales?**
- **How does revenue accumulate over time?**
- **Which products underperform and need attention**



DISCLAIMER & NOTE



DISCLAIMER

This project is created for learning and portfolio purposes only.
The dataset used is sample / publicly available data and does not represent any
real company's confidential information



NOTE

This project is part of my work as a Data Analytics Intern and is shared on
LinkedIn and GitHub to demonstrate my SQL skills and analytical approach
to solving business problems.

RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.

```
SELECT  
    COUNT(order_id) AS total_orders  
FROM  
    orders;
```

Result Grid	
	total_orders
▶	21350



CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES..

```
SELECT  
    ROUND(SUM(order_details.quantity * pizzas.price),  
          2) AS total_sales  
FROM  
    order_details  
    JOIN  
    pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

Result Grid	
	total_sales
▶	817860.05



IDENTIFY THE HIGHEST-PRICED PIZZA.

```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY price DESC
LIMIT 1;
```

Result Grid			Filter Rows
	name	price	
▶	The Greek Pizza	35.95	



IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

```
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

Result Grid			Filter
	size	order_count	
▶	L	18526	
	M	15385	
	S	14137	
	XL	544	
	XXL	28	



LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Result Grid			Filter Rows:
	name	quantity	
►	The Classic Deluxe Pizza	2453	
	The Barbecue Chicken Pizza	2432	
	The Hawaiian Pizza	2422	
	The Pepperoni Pizza	2418	
	The Thai Chicken Pizza	2371	



JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

```
SELECT
```

```
    pizza_types.category,  
    SUM(order_details.quantity) AS quantity
```

```
FROM
```

```
    pizza_types
```

```
    JOIN
```

```
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
```

```
    JOIN
```

```
    order_details ON order_details.pizza_id = pizzas.pizza_id
```

```
GROUP BY pizza_types.category
```

```
ORDER BY quantity DESC;
```

Result Grid			Filter Row
	category	quantity	
▶	Classic	14888	
	Supreme	11987	
	Veggie	11649	
	Chicken	11050	



DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```
SELECT
    HOUR(order_time) AS hours, COUNT(order_id) AS order_count
FROM
    orders
GROUP BY hours;
```

Result Grid			Filter Rows:
	hours	order_count	
▶	11	1231	
	12	2520	
	13	2455	
	14	1472	
	15	1468	
	16	1920	
	17	2336	
	18	2399	
	19	2009	
	20	1642	
	21	1198	
	22	663	
	23	28	
	10	8	



JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.

```
SELECT  
    category, COUNT(name)  
FROM  
    pizza_types  
GROUP BY category;
```

Result Grid			Filter Rows
	category	COUNT(name)	
▶	Chicken	6	
	Classic	8	
	Supreme	9	
	Veggie	9	



GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

```
SELECT
    ROUND(AVG(quantity), 0)
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

Result Grid	
	average_pizzas
▶	138



DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
    JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid			Filter Rows:	
	name	revenue		
▶	The Thai Chicken Pizza	43434.25		
	The Barbecue Chicken Pizza	42768		
	The California Chicken Pizza	41409.5		



CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```
SELECT
    pizza_types.category,
    ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
        ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_sales
    FROM
        order_details
        JOIN
            pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100,
    2) AS revenue
FROM
    pizza_types
    JOIN
        pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
    JOIN
        order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

Result Grid			Filter
	category	revenue	
▶	Classic	26.91	
	Supreme	25.46	
	Chicken	23.96	
	Veggie	23.68	



ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```
select order_date,
sum(revenue) over(order by order_date) as cum_revenue from
(select orders.order_date,
sum(order_details.quantity * pizzas.price) as revenue
from order_details
join pizzas
on pizzas.pizza_id = order_details.pizza_id
join orders
on orders.order_id = order_details.order_id
group by orders.order_date) as sales;
```

Result Grid			Filter Rows:
	order_date	cum_revenue	
▶	2015-01-01	2713.85000000000004	
	2015-01-02	5445.75	
	2015-01-03	8108.15	
	2015-01-04	9863.6	
	2015-01-05	11929.55	
	2015-01-06	14358.5	



DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```
select name , revenue from
(select category, name, revenue,
rank() over(partition by category order by revenue desc) as rn
from
(select pizza_types.category, pizza_types.name,
sum(order_details.quantity * pizzas.price) as revenue
from pizza_types
join pizzas
on pizzas.pizza_type_id = pizza_types.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a ) as b
where rn <= 3;
```

Result Grid			Filter Rows:
	name	revenue	
►	The Thai Chicken Pizza	43434.25	
	The Barbecue Chicken Pizza	42768	
	The California Chicken Pizza	41409.5	
	The Classic Deluxe Pizza	38180.5	
	The Hawaiian Pizza	32273.25	
	The Pepperoni Pizza	30161.75	
	The Spicy Italian Pizza	34831.25	
	The Italian Supreme Pizza	33476.75	
	The Sicilian Pizza	30940.5	
	The Four Cheese Pizza	32265.70000000065	
	The Mexicana Pizza	26780.75	
	The Five Cheese Pizza	26066.5	





THANK YOU